

# The half-cell registration defect

For the whole of round 1, the benchmark compared a low-fidelity field against a high-fidelity field that was sitting half a grid cell away from it. Nothing was noisy, nothing was random, and no optimiser was involved — two pieces of code simply disagreed about where a grid point *is*. This note explains the disagreement, what it cost, what has been repaired, and what has not.

|                         |                         |                           |
|-------------------------|-------------------------|---------------------------|
| FOUND                   | CONFIRMED               | REPAIRED IN               |
| 2026-07-30 · s3_warp-B1 | Closed form, no fitting | round2/eval/ · 2026-07-31 |
| STILL OPEN              |                         |                           |
| Generators · model code |                         |                           |

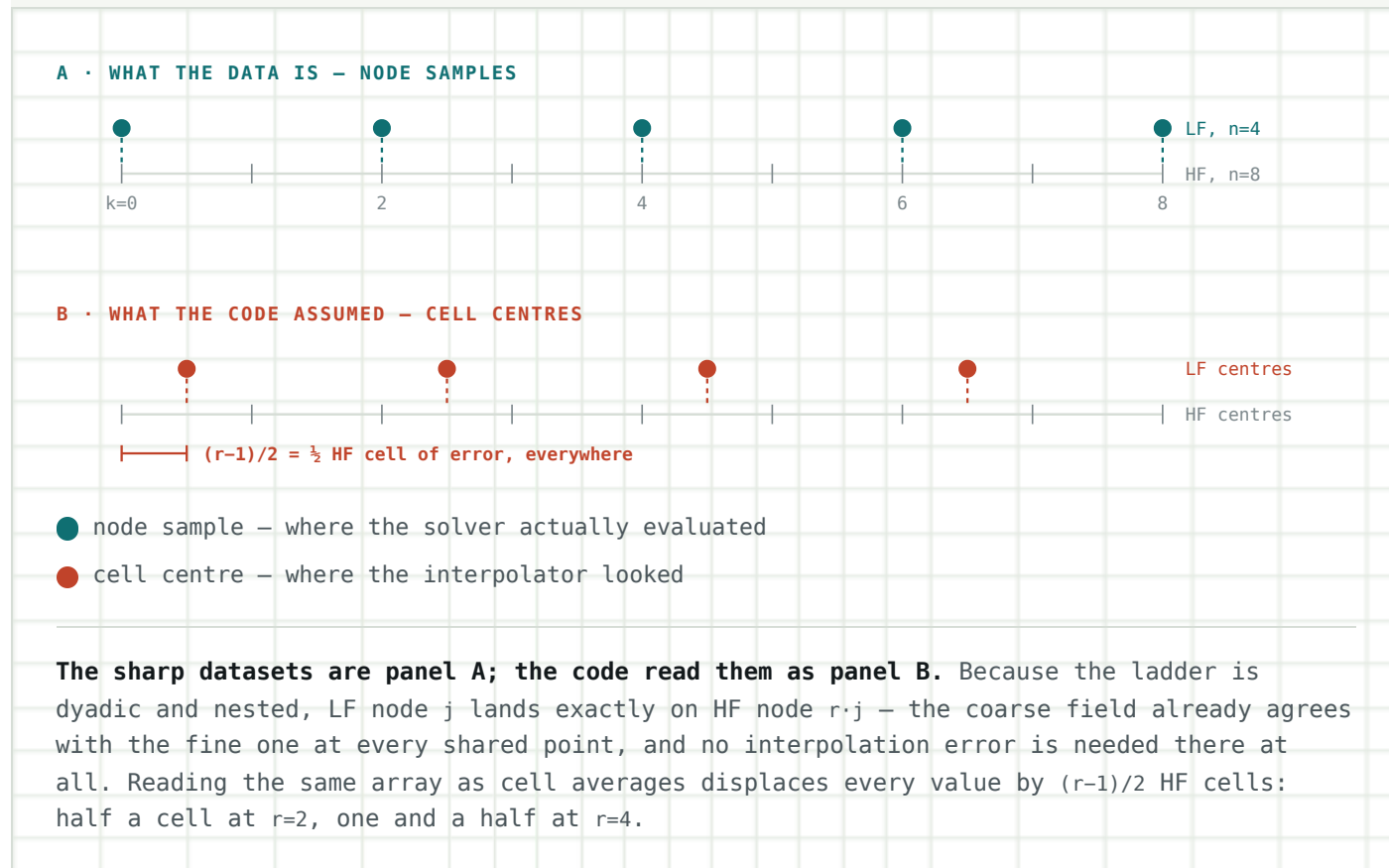
|                                                                                                                                                                                    |                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>REPAIRED</div> <div>The round-2 scoring reference. Copy-LF is now rebuilt from native-grid LF under a per-dataset grid convention, and the construction is hash-sealed.</div> | <div>STILL BROKEN</div> <div>The dataset generators, and every model that upsamples LF itself. The models carry the identical defect — and fixing the reference stopped it from cancelling out.</div> |
| <div>HANDLED BY CAVEAT</div> <div>Two datasets that the corrected reference exposes as degenerate or scale-broken. Documented, not repaired.</div>                                 |                                                                                                                                                                                                       |

## 01 Where a grid point is

A solver stores a field as a list of numbers. What those numbers *mean* is a convention, and there are two common ones.

Under the **node convention**, entry  $j$  is the value of the solution *at a point*:  $x_j = j \cdot L/n$ . This is what a pseudo-spectral solver produces — it works with Fourier coefficients of a function sampled at points. Under the **cell-centred convention**, entry  $j$  is the average over a little box, and the box centre sits at  $x_j = (j + \frac{1}{2}) \cdot L/n$ . This is what a finite-volume solver produces.

The two conventions put the samples in physically different places, and the gap between them depends on the resolution. Halve the grid spacing and the node positions stay put, but the cell centres move. That is the entire bug.



## 02 How we know – a probe with no physics in it

The defect surfaced as an anomaly, not a hunt. The `s3_warp` experiment fitted a displacement field to warp the coarse prediction onto the fine truth. Its answer came back as a near-constant `+0.5` cells on both axes across every sharp dataset — carrying 89–98% of the total displacement energy on two of them. A "warp" that is almost entirely a constant is not transport.

Three properties said *convention* rather than *physics*: the shift was identical in cell units at both  $128^2$  and  $256^2$  (physics would scale with grid spacing, not with the

grid), it was isotropic to within 0.05 cells in 100 of 100 samples, and it vanished on the one dataset from a different generator.

What settled it was a probe containing no PDE and no data. Feed a **linear index ramp** through the real upsampling function. Bilinear interpolation of a linear function is exact, so the output *is* the source coordinate that each output pixel read from. Here is that probe, run at `r=2`, showing which coarse index each fine pixel samples:

sampled LF index at HF rows 0...5

|                                            |                   |                   |                   |                   |                   |                   |
|--------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| <code>scipy zoom(grid_mode=True)</code>    | <code>0.00</code> | <code>0.25</code> | <code>0.75</code> | <code>1.25</code> | <code>1.75</code> | <code>2.25</code> |
| <code>torch F.interpolate(ac=False)</code> | <code>0.00</code> | <code>0.25</code> | <code>0.75</code> | <code>1.25</code> | <code>1.75</code> | <code>2.25</code> |
| <code>torch F.interpolate(ac=True)</code>  | <code>0.00</code> | <code>0.47</code> | <code>0.93</code> | <code>1.40</code> | <code>1.87</code> | <code>2.33</code> |
| <code>node-aligned (k / r)</code>          | <code>0.00</code> | <code>0.50</code> | <code>1.00</code> | <code>1.50</code> | <code>2.00</code> | <code>2.50</code> |

`max | zoom - F.interpolate(ac=False) | = 0.0 ← bit-identical`

The correct answer for nested node data is the last row: fine pixel `k` reads coarse index `k/r`, so every other fine pixel lands exactly on a coarse sample. The eval layer was computing `(k+0.5)/r - 0.5` instead — a quarter of a coarse cell short, which is half a fine cell. Deviation from that formula: `0.0`, on the `128 → 256`, `64 → 128` and `24 → 96` maps alike.

Note the third row. `align_corners=True` is the reflex fix and it is also wrong — it stretches the coarse grid to span the fine grid's endpoints, which is a different error, not the absence of one.

## Four independent confirmations

- The generators' own interpolator satisfies `up[:, :2, :2] == coarse` to `8.9e-16` — node, unambiguously.
- On real test data, `nRMSE(LF, HF[:, :2, :2])` beats `nRMSE(LF, block-mean)` by 2.2–7.7× — and by 150,000× on one dataset. The data votes node.
- Closed-form Lucas–Kanade, sharing no code with the original fitter, measures `(+0.4995, +0.4996)` on `allen_cahn` and `(+0.5020, +0.4889)` on `cahn_hilliard`.
- Helmholtz, the apparent counterexample, is the same defect in disguise: its non-nested Dirichlet interior-node grid at `r=4` predicts a zero-mean  $\pm 1.5$ -cell

*stretch* rather than a constant offset — and the recorded displacement has the predicted sign, the predicted slope to 15%, and the predicted zero mean. A constant-offset statistic structurally cannot see it.

### 03 Why it survived a whole round

Three things had to line up, and all three did.

- **The bug was upstream of the eval layer.** The dataset generator's own helper makes the same cell-centred assumption — and its docstring claims the opposite ("coarse grid points land exactly on HF grid points"), which is only true under the convention it does not use. The eval layer inherited a benchmark-construction error; reading the eval code alone would not reveal it.
- **The self-test could not see it.** The warp rig's validation synthesised its fake coarse field *on the fine grid*. The coarse → fine upsample was therefore never in the tested path — the one operation under suspicion was the one operation the test skipped.
- **The defect partly cancelled.** Models that upsample LF used the same convention as the reference, so both sides were shifted together. A comparison between two equally-wrong things looks healthy. The models were even rewarded for it — one celebrated corrector turned out to be 87–98% explained as having learned the analytic half-cell phase ramp.

The structural lesson is narrow and actionable: *the metric was hashed and sealed; the construction of the reference it divides by was not*. Round 2 closes that specific hole with a separate `copylf_def_hash`, and the loader now refuses to score a dataset that has no grid convention explicitly assigned to it.

### 04 What it cost

Skill is `nRMSE(model) / nRMSE(copy-LF)`. A misregistered copy-LF is an artificially bad denominator, so every skill number in round 1 flattered its model. Correcting the reference — with zero trained parameters, just the right coordinate map — moves the denominator by these factors:

| PANEL DATASET                 | ROUND-1 REFERENCE | CORRECTED | DENOMINATOR INFLATION |
|-------------------------------|-------------------|-----------|-----------------------|
| sharp__allen_cahn_2d          | 0.016152          | 0.001781  | 9.1×                  |
| sharp__phase_field_crystal_2d | 0.044780          | 0.007381  | 6.1×                  |
| sharp__fisher_kpp_2d          | 0.062632          | 0.021450  | 2.9×                  |
| sharp__cahn_hilliard          | 0.087660          | 0.041803  | 2.1×                  |
| ext__helmholtz_2d             | 0.329450          | 0.299033  | 1.1×                  |

Two consequences point in opposite directions, and both matter.

Every **failure** claim in round 1 survives and gets worse — the models were losing to a reference that was itself half a cell out, so the true gap to an honest coarse-solution passthrough is 2–9× larger than recorded. Every **win** claim on the sharp panel has to be re-read: one model's headline result was measured, by direct field projection, at 95.9–100% registration repair, and a fixed zero-parameter resample beat the trained model by 10.3×.

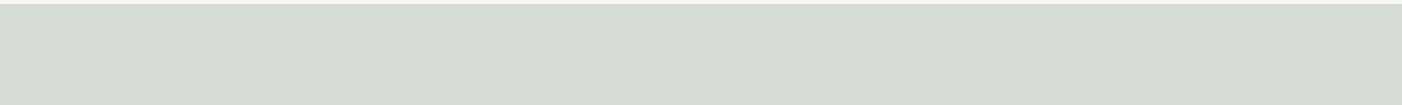
Contrasts *within* a card remain valid throughout — both arms shared the reference, so it cancels exactly. The round-1 metric stays frozen for internal consistency; this is interpretation debt, not a retraction.

EXPOSED BY THE FIX

Under a correct reference, `sharp__phase_field_crystal_2d` has essentially **no fidelity gap at all**. Its solver uses a resolution-independent continuous symbol on a spectrally-resolved state, so the coarse and fine solves are the same solution — the fraction of fine-grid energy outside the coarse band is `4.9e-16`. Its entire round-1 "error" was the interpolation convention. The dataset needs a redesign (a coarser LF rung, or a resolution-dependent symbol) before it can carry a multi-fidelity claim.

05 What has been repaired

All of this landed in `round2/eval/` before round 2 launched, and none of it touches the frozen round-1 numbers.



## Per-dataset convention dispatch DONE

`copylf_prediction` now rebuilds the reference from *native-grid* LF and picks the coordinate map by dataset: node-aligned bilinear with periodic wrap for the nested pseudo-spectral sets, a Dirichlet interior-node map for helmholtz, the legacy map only where an audit certified it consistent. An unclassified dataset raises rather than defaulting — the convention is an explicit decision, not a fallback.

`round2/eval/panel_data.py`

## The wrap seam, folded in DONE

The old call clamped at the domain edge on periodic fields. The replacement wraps. Worth 0.12–0.74% of copy-LF error — real, and two orders of magnitude below the registration shift on the same datasets.

## References and floors re-baselined and sealed DONE

Corrected denominators recomputed with seam checks (datasets that should be unchanged reproduce round 1 bit-for-bit). Floor arms — nearest-condition, train-mean, and zero — recomputed under the corrected reference and frozen before any round-2 experiment ran. Result JSONs now carry the reference-construction hash independently of the metric hash, and cross-hash comparison is refused.

## 06 What still needs fixing

Ranked by how much each one distorts a live number. The first item is the one that the repair itself created exposure to.

### 1 · The models carry the same defect OPEN

Every model-side coarse → fine upsample in the repo calls `F.interpolate(mode="bilinear", align_corners=False)` — verified above to be *bit-identical* to the defective `zoom(grid_mode=True)`. In round 1 this cancelled against an equally-shifted reference. Under round 2's corrected reference it no longer does: a model feeds itself a half-cell-shifted coarse field and is then scored against a correctly registered one, so it must spend capacity learning an inverse phase ramp before it can learn any physics. This lands squarely on the stream studying what signal the coarse solve actually carries.

The fix is not a global flag flip. `align_corners=False` is *correct* for genuinely cell-centred data, and only the sharp panel has been proven node-sampled — so this needs the same per-dataset dispatch the eval layer got, with the index-ramp probe run once per dataset to decide.

`factory_mffp/models/*/smoke_eval.py` · `models/_common/fire_core.py` · `akash/common/mffp.py` · `akash/models/mf_fno_*/model.py`

## 2 · The generators still write shifted arrays OPEN

The dataset builder stores a set of pre-aligned coarse fields alongside the raw ones, and builds them with the defective interpolator — so the defect is baked into the arrays on disk, and into the fidelity-gap metrics the datasets report about themselves. Round-2 eval sidesteps this by rebuilding from raw, but anything that reads the stored aligned arrays inherits it, and any regeneration reintroduces it. This surface is mentor-owned; the note is a recommendation, not an action.

`mffp_sharp/common/ladder.py` – `upsample_to_hf`, `assemble_sample`

## 3 · Two target-scaling defects in the model families OPEN

Surfaced by the same analysis and unrelated to grids. On helmholtz, one training sample of 400 carries 89.2% of the loss energy under the family's single global target scaler — an effective sample size of 1.2; a single train-fitted scalar turns a 4.14 failure into 0.995. On phase-field-crystal the global scaler is 79,000× the median per-sample magnitude, so targets are numerically zero and the model emits a noise floor 8,206× the truth. Round 2 has the eligibility *rule* for per-sample normalisation; these two concrete defects are unrepaired.

## 4 · Phase-field-crystal needs a redesign CAVEATED

Currently handled by a mandatory caveat on every claim rather than a fix. Under the corrected reference it has no fidelity gap to predict; it should get a coarser LF rung or a resolution-dependent symbol, or be moved to report-only.

## 5 · Two deferred plumbing repairs DEFERRED

The corrector's *input* wrap seam (deliberately excluded twice, because fixing it mid-round would have confounded a paired contrast) and permuting queries before chunking in the attention family (recovers the unchunked forward pass to 0.014% at zero cost). Both are small; both are still owed.

## 07 The check, for any future dataset

This class of bug is invisible in every aggregate metric and takes about a minute to rule out. The recipe generalises to any resampling code in any repo:

1. Push a **linear index ramp** through the resampler. The output is literally the source coordinate each output element read.
2. Compare against `k/r`. Any constant difference is a registration offset; any slope is a stretch.

3. Cross-check against the data itself: is the coarse field closer to a decimation of the fine field, or to a block average of it? The winner names the convention.
4. Assert the invariant in a test. For nested node data: `upsample(coarse)[::r, ::r] == coarse`, exactly.

Step 4 is the one that would have caught this on day one — and it is one line.

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Sources – s3\_warp-B1 mechanism analysis (parts 5–6) and its turn-1 reanalysis; round-1 report §5; the mentor-facing registration note of 2026-07-30 and its two addenda; round-2 program §2.2–2.3 and launch gates G1/G2/G3-r2. The `F.interpolate` comparison in §02 was run fresh for this note. Round-1 scores remain frozen; nothing here alters a recorded number.